SIOC 251: Radiative Transfer

Homework 1: Monte Carlo Model of Solar Radiation in an Absorbing Atmosphere

1 Overview

In this assignment you will build your first radiative transfer model: a simple Monte Carlo simulation of photon extinction in a homogeneous atmosphere. The goals are to:

- develop intuition about optical depth, extinction, and transmittance
- gain experience implementing Monte Carlo models
- compare numerical simulations with analytical radiative transfer results
- interpret real atmospheric measurements using AERONET data

Your atmosphere will initially contain only absorption (no scattering). Later assignments will extend this model to include scattering and clouds or aerosols.

If you would like additional background on Monte Carlo radiative transfer methods, a useful reference (mainly chapter 2) is available on the course Canvas site:

`https://canvas.ucsd.edu/courses/75371/files?preview=17602295`

2 Repository Structure and Submission

Your work should be added to the course repository in the following structure:

```
sioc251_radiative_transfer
|
|-- README.md
|-- environment.yml
|
|-- HW0_prism_dispersion
|
|-- HW1_monte_carlo_extinction
       |
       |-- monte_carlo_1.ipynb
       |-- (optional additional notebooks)
       |
       |-- figures
            |
            |-- figure1.png
```

Notes:

- You may develop code in any language, but the final submission must be a **Python notebook**.
- All figures should be saved in the `figures/` directory.
- Push the completed homework 1 folder to your course GitHub repository (same procedure as Homework 0).

3 AI Use Policy

Students are encouraged to use AI tools (e.g., ChatGPT, Claude, Gemini) to assist with coding, debugging, plotting, and documentation. However, students remain fully responsible for the scientific correctness and physical interpretation of their work. AI output should be treated like information from other sources: it must be understood, verified, and appropriately documented.

Policy details

1. Each assignment must include a short **AI Usage** statement describing how AI tools were used.
   Example:
   ```
   Tool: ChatGPT
   Prompt: \Write Python code to generate random numbers from an
   exponential distribution."
   How it was used: Provided a photon pathlength generator
   for the Monte Carlo model.
   ```
2. Students must be able to explain how their code works and how the physics is implemented. AI-generated code that is not understood or validated is not acceptable.
3. Each assignment must include **two model verification tests** demonstrating that the model behaves physically correctly.
4. Students may discuss ideas and debugging strategies with classmates, but submitted code must represent their own work.

4 Part I: Building the Monte Carlo Model

Construct a simple model of a homogeneous atmosphere consisting of a single layer with geometric thickness equal to one atmospheric scale height. Photons enter the top of the atmosphere and travel downward. Each photon either 1) reaches the surface, or 2) undergoes an extinction event in the atmosphere. The key step is determining how far a photon travels before extinction.

The probability that a photon travels optical depth τ without extinction is

$$P(\tau) = e^{-\tau}.$$

The cumulative distribution function is therefore

$$\mathrm{CDF}(\tau) = 1 - e^{-\tau}.$$

If ζ is a uniformly distributed random number between 0 and 1, the sampled optical depth traveled by the photon is

$$\tau = -\ln(1-\zeta).$$

The corresponding geometric pathlength is

$$L = Z_{\mathrm{atm}}\,\mu \frac{\tau}{\tau^*}$$

where Z_{atm} is the atmospheric thickness, $\mu = \cos\theta$ is the cosine of the solar zenith angle, and τ^* is the total optical depth of the atmosphere. Your model should simulate many photons entering the atmosphere and determine whether they reach the surface or undergo extinction.

5 Part II: Model Verification

Before using your model for experiments, you must demonstrate that it behaves physically correctly. Design and perform **two verification tests**. These tests should confirm that the model produces results consistent with known physical behavior. One example would be correct behavior in limiting cases.

For each verification test:

- clearly state the expected physical behavior
- show results from your model
- explain whether the model behaves as expected

6 Part III: Radiative Transfer Experiments

Use both your Monte Carlo model and analytical radiative transfer expressions to answer the following questions.

For Monte Carlo simulations, use between 10^4 and 10^6 photons depending on computational speed.

1. For the sun directly overhead ($\theta = 0$), what is the value of optical depth τ that results in an e-folding reduction in surface intensity?
2. For $\tau = 1$, what is the surface transmittance t when the solar zenith angle is $60°$?
3. Explain why, in a purely absorbing atmosphere, the surface would be dark when the sun reaches the horizon (assume the sun is a point source).
4. For $\tau = 1, 3,$ and 10, how does transmittance vary with height in the atmosphere?

7 Part IV: AERONET Observations

Use real atmospheric measurements to interpret the radiative transfer results.

1. Visit the NASA AERONET website:

   `https://aeronet.gsfc.nasa.gov/`

   Choose a station in a region you find interesting.
2. Download **Level 2.0 AOD data** for a day with significant aerosol variability.
   Satellite imagery from NASA Worldview can help identify interesting events:

   `https://worldview.earthdata.nasa.gov/`
3. Save a plot of AOD for your selected day.
4. Download the corresponding dataset and extract
   - 500 nm (or 550 nm) AOD
   - solar zenith angle
5. Answer the following questions, using your Monte Carlo model where appropriate:
   - What is the primary source of aerosols at your station?
   - How does the direct transmittance at the surface change throughout the day?
   - For a constant τ, which should be the daily mean value, how does the direct surface transmittance change throughout the day?

- For a constant θ, which should be the day's maximum value, how does the direct surface transmittance change throughout the day?
- What factor is most responsible for the estimated changes in transmittance at the surface, τ or θ?

8 Part V: Student Investigation

Design a simple experiment using your Monte Carlo model.

- Pose a question or hypothesis related to atmospheric extinction or transmittance. This can be theoretical only or based on AERONET measurements.
- Use your model to test the idea.
- Present results using figures or calculations.
- Briefly interpret the results.

The question can/should be very simple. In fact, sometimes interesting insights or followup questions come from exploring ideas that initially seem obvious.

Clearly state:

- the question and hypothesis
- the experiment performed
- the result